Technology-Based Early Language Comprehension Intervention (TELCI): Making the Case for Language Comprehension

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TeLCI Team













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Educational Problem

- Approximately 31% of 4th graders read below a basic proficiency level (NAEP, 2015)—that is, they **fail to make simple inferences and understand the overall meaning of texts.**
- For many of these children, a focus on decoding is likely *necessary*, but *insufficient*, and **early instruction in language comprehension skills is also needed to prevent long-term negative reading outcomes** (Foorman et al., 2016)**.**
- Inference making is a core language comprehension skill (Connor et al., 2013, 2014)

Inference Making

- An *inference* is information **retrieved to fill in information** that is left implicit in a text (McNamara & Magliano, 2009).
- Inference making is **the underlying process that supports the construction of coherent representations** (e.g., Kintsch & van Dijk, 1978) in both reading and non-reading contexts (e.g., Gernsbacher, 1990; Kendeou et al., 2008, 2009; Kim 2016; Magliano et al., 2013).
- The ability to draw inferences is ***a general skill***–it is not specific to reading (Kendeou, 2015)

Inference Making

- *Two-stage process*: **activation** of previously acquired information; its **integration** with current content (Oakhill, 1984).
- ***Questions*** can prompt inferences because they: (a) 'cue' activation of relevant information, and (b) facilitate integration of that information – especially when accompanied with ***scaffolding*** and ***immediate, specific feedback*** (e.g., McMaster et al., 2014).
- An important consideration is whether it is more beneficial to prompt the students ***during*** **(i.e., *online*)** or ***after*** **(i.e., *offline*)** the comprehension task.

TeLCI is designed to improve reading comprehension by developing **inference making** in Grades 1-2. TeLCI does not rely on decoding skills, and takes the form of an interactive, cloud-based software application with twenty-four 30-min **learning modules**, each of which engages students to:

- view age-appropriate ***videos*** (half will be *fiction* and half will be *nonfiction genre*),
- learn academic ***vocabulary*** words that are central to the main ideas in the video,
- respond interactively with the software (by touch screen selections) to ***inferential questions*** *online* (during viewing) or *offline* (after viewing),
- receive ***scaffolding*** and specific ***feedback*** for each inferential question.



Project TeLCI

CEHD | College of Education + Human Development

Students with or at risk for disabilities who experience comprehension difficulties

Grades 1-2

TeLCI

- **Key Vocabulary**
- **Fiction and Nonfiction video genre**
- **Scaffolding**
- **Inferential Questioning *(online vs. offline)***
- **Feedback**
- **Transfer**

Usable & Feasible School Implementation with Fidelity

Inference Skills
in non-reading contexts

(the underlying process targeted by TELCI)

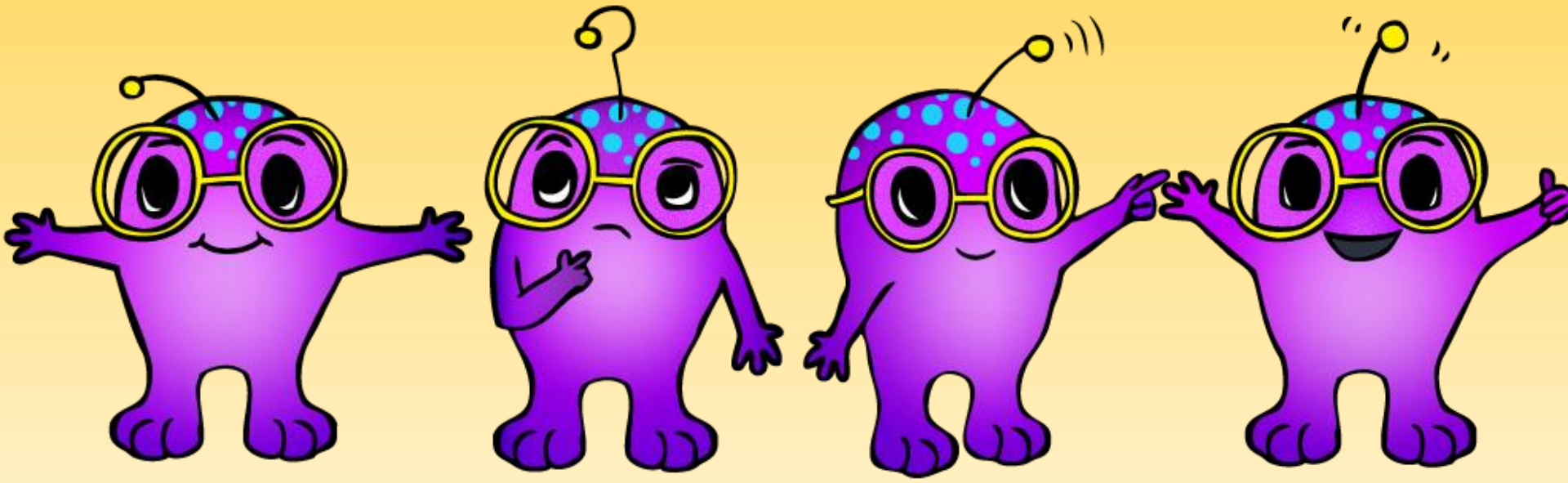
Vocabulary
Decoding
EF

Reading Comprehension
Performance

Research Plan

- **Year 1:** we will develop the **TeLCI software and learning modules**, conduct extensive lab testing and a brief **Field Test of its usability and feasibility**, and revise/refine as needed.
- **Year 2**: we will continue **learning module development**, draft **transfer modules**, a manual, teacher training materials, and **fidelity tools**; examine two versions of the instruction (*online* vs. *offline* questioning) in an **Experimental Field Trial**; and identify the most feasible and promising version(s).
- **Year 3**: we will conduct a Pilot Study to determine the **promise** of the intervention compared to controls.

TeLCI Avatar



TeLCI Modules and Assessments

- Video Selection Parameters:
  - (a) *theoretical soundness* (causal structure)
  - (b) *age-appropriateness*
  - (c) *cultural relevance*
  - (d) likelihood that students are *not familiar* with the videos; and
  - (e) *fit* with district standards, curriculum, and instructional programming

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Field Test

- Participants
  - *N* = 95 students in Grades 1 and 2

  52% female, 54% White, 19% Black, 21% Hispanic

  47% reduced/free lunch, 23% ELL
  - *N* = 8 Teachers; *N* = 2 Literacy Experts
- Methods
  - Focus Groups (Teachers, Literacy Experts)
  - Students Scores (CELF, QRI, MEFS, TeLCI Assessment, TeLCI module data)
  - Usability and Feasibility Observation Questionnaires

Preliminary Results

- Students like interacting with *TeLCI Avatar* and find the modules to be fun and engaging
- *TeLCI Innovative Assessments* work well and have potential for scale-up – psychometric properties to be established
- *TeLCI modules* have been reviewed and approved *by research and school personnel*
- *TeLCI Modules* – fiction and nonfiction are currently being tested across all skill levels

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Thank You

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